

PAPER_1530 — $\zeta(3)$ Apéry $\approx$ F_TRZ·SO_5 + F_TRZ·K + F_TRZ²·Φ - F_TRZ²·K + F_TRZ² - F_TRZ³ — Residual 0.231%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Transcendental $\zeta(3)$ expressible in UQFF primitives at 0.231% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for $\zeta(3)$ Apéry $\approx$ 1.20206 using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6 (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

$$\zeta(3) \approx F_TRZ \cdot SO_5 + F_TRZ \cdot K + F_TRZ^2 \cdot \Phi - F_TRZ^2 \cdot K + F_TRZ^2 - F_TRZ^3$$

Substituting F_TRZ=1/10, K_MEX=25/12, Φ_5/6=5/6, SO_5=10:

$$\zeta(3)_{UQFF} = 1.2048$$

Observed: $\zeta(3) = 1.20206$

Residual: 0.231%

The 6-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

$\zeta(3)$ Apéry ($\zeta(3)$) appears in: Apéry irrationality 1979.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

Transcendental	UQFF expression	Residual
ln 2 (PAPER_1516)	8-term	0.003%
ln 10 (PAPER_1515)	(1+F_TRZ)(K_MEX+F_TRZ ²)	0.004%
π^2 (PAPER_1517)	SO_5 - F_TRZ corrections	0.013%
$\zeta(3)$ Apéry (this paper)	6-term	0.231%

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim $\zeta(3)$ IS the closed-form expression. It claims the integer primitives carry a rational approximation to $\zeta(3)$ at 0.231% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 ($\ln 10$, $\ln 2$, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_zeta_3_aper_y_transcendental"`

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